

2024 AMC 10A Problem 7 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 7. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 7

Problem:

The product of three integers is 60. What is the least possible positive sum of the three integers?

Answer Choices:

- A. 2
- B. 3
- C. 5
- D. 6
- E. 13

Solution:

Note that $60 = 10 \cdot (-1) \cdot (-6)$, and the sum of these factors is (B) 3. It remains to show that no positive sum can be less than 3. Such a sum would have to consist of one positive integer and two negative integers with smaller absolute value. If the positive integer is greater than or equal to 10, then the sum is greater than or equal to 3. The possible sets of factors in this case are $\{10, -6, -1\}$, $\{10, -3, -2\}$, $\{12, -5, -1\}$, $\{15, -4, -1\}$, $\{15, -2, -2\}$, $\{20, -3, -1\}$, $\{30, -2, -1\}$, and $\{60, -1, -1\}$. None of these sets of factors has a sum less than 3.

The only other possible choices for the positive integer are 5 and 6, and in neither case is a positive sum possible. Indeed, if the positive integer is 5, then the only possible set of factors is $\{5, -3, -4\}$. If the positive integer is 6, then the only possible set of factors is $\{6, -5, -2\}$. In both of these

cases, the sum is not positive.

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